

Algebra II with Trigonometry

Chapter 5.4

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4401270_alg-2trig-h-chp-5-4-note-docx.pdf](#)

Certainly! Here are **10 mental math questions** that test the core concepts from the provided PDF on inverse trigonometric functions and right triangles. They are designed to be answerable without a calculator, paper, or pencil, focusing on properties, domains, basic values, and definitions:

1. Domain & Range Understanding

Q1: What is the domain of the inverse sine function, $\sin^{-1} x$?

- **A:** $[-1, 1]$

Q2: What is the range of the inverse cosine function, $\cos^{-1} x$?

- **A:** $[0, \pi]$
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2. Standard Values

Q3: What is $\sin^{-1}(0)$ in radians?

- **A:** 0

Q4: What is $\cos^{-1}(1)$ in radians?

- **A:** 0

Q5: What is $\tan^{-1}(1)$ in radians?

- **A:** $\frac{\pi}{4}$
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3. Connections & Properties

Q6: True or False: $\sin^{-1}(-x) = -\sin^{-1}(x)$ for every x in the domain.

- **A:** True

Q7: If $\theta = \arccos(0)$, what is θ in radians?

- **A:** $\frac{\pi}{2}$
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4. Right Triangle Reasoning

Q8: If the opposite and adjacent sides of a right triangle are equal, what is $\arctan(1)$ in degrees?

- **A:** 45°

Q9: If $\sin \theta = 1$, what is θ in radians (principal value for inverse sine)?

- **A:** $\frac{\pi}{2}$
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5. Composite Function Intuition

Q10: If x is in the range $[-1, 1]$, and you calculate $\sin(\sin^{-1}(x))$, what do you get?

- **A:** x
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Feel free to ask for more challenging or applied versions!